

THE DIGITAL AGRICULTURE MATURITY MODEL

A Guidebook

Version 1.7 · Draft for review · August 2026

Prepared as a companion to the Digital Agriculture Roadmap (DAR) Playbook

Contents

Abbreviations.....	3
Executive Summary.....	3
1. Introduction and Purpose.....	5
1.1 The role of the model, precisely stated.....	5
1.2 The case against maturity models, and the design answer.....	5
1.3 The version 1.7 package.....	6
2. What the Model Owes Its Sources.....	6
3. What the Model Measures.....	8
3.1 Seven pillars.....	8
3.2 Four layers and six use case areas.....	8
4. Rules of Evidence.....	9
4.1 No level without a recorded value.....	9
4.2 Four evidence classes, derived rather than declared.....	9
4.3 Staleness.....	10
4.4 One ladder for non-measured indicators.....	10
4.5 Reading a band on weak evidence.....	10
5. Prerequisites.....	11
6. The Assessment Process.....	11
Step 1: Assemble the evidence.....	12
Step 2: Derive the scores.....	12
Step 3: Render the diagnostic report.....	13
Step 4: Use the report in roadmap preparation.....	13
7. Reading the Diagnostic Report.....	13
7.1 Reading the visuals.....	14
8. What the Model Will Not Do.....	16
9. Country Applications.....	16
10. Limits of the Current Version.....	17
Annex A. The Indicator Register.....	19
Annex B. Scoring Reference.....	23
B.1 Bands.....	23
B.2 The ladder.....	23
B.3 Rules and constants, with their meanings.....	23
Annex C. Version History.....	24

Abbreviations

DAMM	Digital Agriculture Maturity Model
DAR	Digital Agriculture Roadmap
DPI	digital public infrastructure
EGDI	United Nations E-Government Development Index
GCI	Global Cybersecurity Index (ITU)
GII	Global Innovation Index (WIPO)
KPI	key performance indicator
NDPA	Nigeria Data Protection Act
NIN	National Identification Number (Nigeria)
TTL	task team leader
WDI	World Development Indicators (World Bank)

Executive Summary

A Digital Agriculture Roadmap (DAR) is a strategy, investment, and implementation plan for transforming a country’s agriculture sector with digital technologies, prepared under the five-step method of the DAR Playbook (World Bank, Gates Foundation, and BCG 2025). Every step after the first depends on the quality of the diagnostic. The Digital Agriculture Maturity Model (DAMM) exists to produce that diagnostic, and nothing else: it executes the Playbook’s maturity assessment (step 1C), supplies the readiness and impact evidence for use case prioritization (step 2B), and hands the results framework a measured baseline (step 3C).

Version 1.7 observes a country through 57 indicators in seven pillars, cut from 102 in earlier versions by requiring every indicator to name the roadmap decision it serves. Thirty-two indicators score against numeric thresholds and 25 against a single shared capability ladder; 19 draw on global series a machine can fetch, and 38 are country work. Four rules carry the method: no level may exist without a recorded value; the evidence class of every reading (Measured, Documented, Judged, or Gap) is derived from what was recorded, never declared; readings older than three years are flagged stale; and eleven prerequisites bind on presence only, each blocking a specific use case rather than capping an overall result. There is no overall score and no stage, by design.

The process has four steps, of which two belong to people and two are automatic. The instrument is a family of three workbook files, a blank template and two worked examples (Arab Republic of Egypt and Nigeria), each carrying the full scoring register, a computed dashboard, and a Visuals sheet of live charts. The endpoint is a single diagnostic report per country in which the indicators, their evidence, the visualizations, and the diagnostic narrative travel together.

The model was assessed against two countries in August 2026, both by the same process. On Egypt it ordered priorities and dated the evidence — and then declined to call five use case columns ready, because the one figure that would settle them, rural mobile broadband coverage, is not published for

Egypt at any admissible tier. On Nigeria it produced a legible structure on evidence that a shallower pass had treated as missing, and capped every column at partial on a single foundation, rural electrification at 23.5 percent. In both it converted what is not known into a short, costed verification agenda rather than a false score. That second behavior is the one earlier versions could not produce.

This guidebook is a draft for review. Section 10 states the limits of the current version candidly; four standing prohibitions, beginning with a bar on cross-country ranking, apply to every output.

1. Introduction and Purpose

Digital technologies are changing how agricultural services reach farmers: advisory that once traveled through an extension visit now travels through a phone, credit decisions that once required collateral can draw on transaction histories, and market prices that once favoured the best-connected trader are increasingly public. The gains, however, have been uneven. More than 150 million smallholder farmers in developing countries still lack access to digital services, and fewer than 10 percent of potential users in low- and middle-income countries use digital agriculture tools (World Bank, Gates Foundation, and BCG 2025).

A Digital Agriculture Roadmap (DAR) is a strategy, investment, and implementation plan for a country or region to guide transformation of its agriculture sector using digital technologies. The DAR Playbook (World Bank, Gates Foundation, and BCG 2025) sets out a five-step method for preparing one: diagnostic assessment, use case prioritization, vision development, roadmap development, and implementation planning. Every step after the first depends on the quality of the diagnostic. That diagnostic is what the Digital Agriculture Maturity Model (DAMM) produces.

This guidebook documents version 1.7 of the model. It describes what the model measures, the rules of evidence it enforces, the four-step process for applying it, and the diagnostic report that is its single output. It is written for task team leaders (TTLs), country teams, and partners who commission or conduct assessments, and it assumes no familiarity with earlier versions.

1.1 The role of the model, precisely stated

DAMM serves three specific points in the Playbook method and no others. It executes step 1C, the determination of ecosystem maturity. It supplies the digital readiness and impact evidence that step 2B uses to prioritize use cases. And it hands step 3C a monitoring baseline, since every measured reading carries a value, a source, and a year. The model does not draft the roadmap, does not select use cases, and does not make investment decisions. Those are tasks for the team and the government, informed by the diagnostic but never determined by it.

1.2 The case against maturity models, and the design answer

The case against maturity models deserves its strongest form. Critics point out that (a) composite scores invite league tables, whatever the accompanying caveats say; (b) staged models compress dozens of separate judgments into one number that hides more than it reveals; (c) diagnostic frameworks tend to grow, adding indicators faster than any country team can populate them; and (d) frameworks routinely present expert opinion in the visual language of measurement, so that a judgment and a data point become indistinguishable on the page.

Version 1.7 accepts each of these criticisms and is built around them. It produces no overall score and no stage. It derives the evidence class of every reading from what was actually recorded, so a judgment can never wear the costume of a measurement. It was cut from 102 indicators to 57 by requiring every indicator to name the roadmap decision it serves. And it is subject to four standing prohibitions, set out in section 8, of which the first is that no cross-country ranking may ever be produced from its outputs.

1.3 The version 1.7 package

Table 1.1 The package: what each component is for

Component	Role
This guidebook	The method, the rules of evidence, the process, and how to read the outputs
Scoring workbook — Blank Template	The instrument for a new country: register, dashboard, and Visuals sheet
Scoring workbooks — Egypt; Nigeria	Worked examples of the same instrument, populated by the August 2026 test runs
Diagnostic reports — Egypt; Nigeria	The endpoint: indicators, evidence, visualizations, and narrative in one document per country
Specification and indicator census	The design record: every keep, merge, and drop decision from 102 to 57, with the evidence behind it
Method companions — source tiers, quality control, practice library	The rules an assessment follows: what each source tier may carry, the three quality gates and the issues log, and the standing library behind any precedent pointer

2. What the Model Owes Its Sources

Version 1.7 was not designed on a blank page. Its indicators and its scoring rules are, in large part, lessons taken from three bodies of work: the DAR Playbook, which defines the roadmap the model serves; the SAID maturity model, the 2020 analysis from which the DAR approach descends; and the Bhutan assessment of March 2026, the most complete country application of the model's earlier versions. It is worth stating precisely what each contributed, both because the debts are real and because knowing why a rule exists is the best protection against amending it casually.

The DAR Playbook (World Bank, Gates Foundation, and BCG 2025). The Playbook's deepest contribution is the selection test that cut the register from 102 indicators to 57. An indicator survived only by naming the use case area and the Playbook step it serves; indicators that could name neither were retired, and the census records each decision. The Playbook also supplied the taxonomy those tags are written in, the six use case areas of its steps 1B and 2A, and it reshaped the outputs: scoring produces a per-use-case readiness matrix because step 2B asks for digital readiness per use case, an impact-side reading from the agriculture pillar because step 2B's other criterion is impact, and a sourced KPI baseline because step 3C needs one. Lastly, the Playbook's own advice that a lighter-touch diagnostic is preferable where evidence already exists gave the warrant for deleting machinery: a diagnostic that serves a five-step method has no need of stages, composite scores, or weights of its own.

The SAID maturity model (BCG for the Gates Foundation, 2020). SAID assessed more than 50 indicators across five categories for 14 geographies and found that the indicators which meaningfully differentiated them were few and highly correlated with one another. That finding is the intellectual basis of the version 1.7 merge rule: where two indicators moved together in both worked examples, they were merged, and the evidence for each merge is recorded in the census.

Three further inheritances are visible in the register. SAID's five categories anticipate the pillar structure, and its finding that business enablement separates geographies is why general enablement indices remain in a digital agriculture model. Its observation of a step change between geographies lacking core digital infrastructure and all others, with foundational investment needed before digital agriculture interventions can matter, is the empirical ancestor of the three universal prerequisites. And its two documented pathways to a strong ecosystem, government-led and private-sector-led, are why the innovation pillar measures public delivery mechanisms and the venture ecosystem separately rather than as one. One SAID device was deliberately not inherited: its four-stage segmentation, built to prioritize a portfolio across 14 geographies, serves a ranking purpose that the first prohibition forbids for a single-country diagnostic.

The Bhutan assessment (March 2026). Bhutan is the proof of feasibility behind the strictest rule in the model. Its assessment populated 97 of 97 indicators with a value, a source, a year, and a confidence rating, with no blanks; the mandatory-value rule of section 4.1 asks future assessors for nothing that has not already been done. Bhutan also demonstrated that indicators labelled qualitative can carry recorded evidence, scoring numeric values on 20 of 42 such indicators, which pointed toward deriving the evidence class from the record rather than asking for an opinion about it. Its four-layer structure, Foundation through Outcomes, is restored in version 1.7, and with it the structural gap computation that earlier versions defined but could not compute. Its practice of scoring the agriculture pillar and letting it drive investment priorities is the precedent for pillar A1. Two of its choices serve as cautions rather than precedents: expert scores expressed on a 0 to 100 scale dressed judgment in the costume of measurement, which the derived classes now prevent, and its single overall score of 2.29 is precisely the number version 1.7 declines to produce.

Table 2.1 What the indicators and the scoring take from each source

Source	What the indicators take	What the scoring takes
DAR Playbook (2025)	The selection test (name your use case and step, or leave the register); the six use case area tags on every indicator	The shape of the outputs: readiness matrix for step 2B, impact reading from A1, KPI baseline for step 3C; the warrant for removing stages, composites, and weights
SAID (2020)	The redundancy principle behind the merges; retention of business-enablement indices; measuring public and private pathways separately	The universal prerequisites, from the step change SAID observed below a floor of core infrastructure; the refusal of stage segmentation for single-country work
Bhutan (2026)	Proof that all 57 can carry value, source, and year; the precedent for scoring agriculture as a pillar	The mandatory-value rule; derived evidence classes; the restored four layers and the computable structural gap; the caution against 0–100 expert scores and a single overall number

A fourth source entered in August 2026 as a counter-example: two AI-generated desk diagnostics of Egypt and Nigeria (You.com ARI), examined side by side with the model. The reports are fluent, broad, and fast, with 150 to 200 references each; they also have no source hierarchy, citations that resolve to domain roots rather than documents, no recorded gaps, and no separation between what was measured and what was asserted. The comparison changed the package rather than the scoring.

It motivated the source-tier protocol (tiers T1 to T5, reported beside each entry and never weighted), the initiative and solutions register in the report, and the firewall rule that researched narrative may cite scored rows but never set a level.

3. What the Model Measures

The model observes a country through 57 indicators organized in seven pillars. Each indicator also carries three further attributes: the layer it belongs to, the use case areas it bears on, and, for eleven of them, a prerequisite role. The full register is at annex A.

3.1 Seven pillars

Table 3.1 The seven pillars

Pillar	What it covers	Indicators	Interpretation
A1	Agriculture and need: productivity, losses, market and credit access, food security, women's landholding	10	The problem digital must answer. A low band is a large opportunity, not a poor performance.
C1	Connectivity and access: rural coverage, affordability, devices, electricity	6	Digital capability
C2	Data and DPI: registries, land records, open agricultural data, weather, soil, interoperability	8	Digital capability
C3	Policy and safeguards: data protection, cybersecurity, strategies, digital ID, coordination	8	Digital capability
C4	People and institutions: literacy, farmer digital skills, extension capability, ministry capacity, gender balance	7	Digital capability
E1	Innovation, solutions, and emerging technology: venture ecosystem, platforms in market, fintech rails, AI deployment and safeguards	11	Ecosystem and solutions
O1	Outcomes and inclusion: adoption by smallholders, women, and youth; financial inclusion; documented impact	7	Results

Agriculture is scored, not annexed. Earlier versions collected the agricultural context and then excluded it from every calculation. Version 1.7 scores it as pillar A1 and reads it as the statement of need: it writes the first chapter of the roadmap and the impact side of use case prioritization. This corrects the defect of a digital agriculture diagnostic that measured the sector it exists for and then set the measurement aside.

3.2 Four layers and six use case areas

Independently of pillar, every indicator sits in one of four layers: foundation (16 indicators), enablers (26), transformation (8), and outcomes (7). The layers are a reading lens, never an aggregation with

weights. Their one computed product is the structural gap between foundation and transformation, described at annex B, which flags either services running ahead of their rails or foundations lying unused.

Every indicator also names the use case areas it bears on, drawn from the Playbook taxonomy: advisory and extension, smart farming, market linkage, supply chain, financial services, and agricultural intelligence. This tagging is what allows the diagnostic to answer the question step 2B actually asks, which is not "how mature is the country" but "which use cases is this country ready for."

Box 2 · Why there is no overall score

A single maturity number would be convenient, and every reviewer of an early draft asked for one. It is refused for two reasons. First, a composite invites the league table that the first prohibition forbids: two numbers will be ranked, whatever the accompanying text says. Second, it destroys the information the roadmap needs. Egypt's profile, foundations Advanced, outcomes Emerging, is a finding a team can act on; compressed to one number it becomes a grade that flatters the rails and hides the farmers. The executive read-out of a DAMM assessment is one sentence built from the pillar profile, and section 7 shows that a sentence is enough.

4. Rules of Evidence

The rules in this section are the substance of version 1.7. They are few, and none is optional.

4.1 No level without a recorded value

Every indicator must carry a value, a source, and an observation year before it may carry a level. The value may be a number, a citation to a specific artifact (a law in force, a registry in operation), or an explicit record that the data point was searched for and not found. What is not permitted is a level resting on an empty cell. In the version 1.5 Egypt assessment, 64 of 102 indicators carried a level with no recorded value; under version 1.7 that assessment could not have been saved.

4.2 Four evidence classes, derived rather than declared

Table 4.1 The four evidence classes

Class	What was recorded	How it scores
Measured	A number, with source and year	Against the indicator's numeric thresholds
Documented	A citable artifact from an admissible source: a law, a registry, a named platform	Against the shared ladder (section 4.4)
Judged	An assessor statement with no artifact behind it	Ladder, visibly flagged as judgment
Gap	A search that found nothing, with the search recorded	No level; feeds the verify list

The class is derived from the content of the value field, never chosen from a menu. This replaces the confidence tags and confidence weights of earlier versions, which recorded the assessor’s opinion of their own evidence and could contradict it. A derived class cannot contradict the evidence because it is computed from the evidence. Admissibility is part of the derivation: a citation whose only source is tier 5 — press, vendor or market material — derives as judgment, not documentation, because tier 5 may never carry a scored value.

Box 1 · A gap that has been looked for and named is not a blank

The model distinguishes a recorded gap from an unassessed indicator. A recorded gap is a finding: someone searched named sources on a stated date and found nothing. It carries no level and no weight, but it appears in the diagnostic report’s verify list, where it becomes a line item in mission planning. In the August 2026 assessments, the four recorded gaps that survived a second, adversarial search in each country were the most useful line items the exercise produced: three of the four are the same indicators in both countries, which says more about what the sector measures than about either country.

4.3 Staleness

A reading whose observation year is more than three years old is flagged stale. One rule covers all 57 indicators; the per-indicator age limits of earlier versions are retired. Stale readings keep their levels but appear on the refresh list, and the diagnostic report displays the vintage of every reading so that the age structure of the evidence is visible at a glance. The Egypt assessment carries 11 stale readings, several of them — the sex-disaggregated landholding count from the 2009/10 agricultural census among them — on exactly the indicators where the inclusion questions lie.

4.4 One ladder for non-measured indicators

Indicators without numeric thresholds (25 of the 57) are scored on a single shared ladder, asked in stages. First, presence: is the capability absent, announced, adopted, or operating? This is a factual question with a citable answer. Second, for capabilities that are at least adopted, quality: ownership, funding, and institutionalization. Third, for capabilities that are operating, scale: coverage and reach against the indicator’s own scale measure. The level derives mechanically: absent scores 1, announced 2, adopted 3, operating with quality 4, and operating with quality, scale, and evidence of use 5.

The ladder replaces the per-indicator anchor paragraphs of version 1.5, of which 35 of 44 were a single template with the indicator name substituted in. Nine genuinely bespoke anchor sets survive as indicator-specific guidance on what counts as quality and scale, and extending that guidance, prerequisites first, is the standing improvement path.

4.5 Reading a band on weak evidence

A pillar band is the unweighted mean of its indicators’ levels, on non-overlapping bands stated at annex B. Where judged and gap readings outnumber measured and documented ones, the band prints in parentheses and should be read as a hypothesis with a work plan attached, not as a finding. This majority rule replaces the coverage percentages and adequacy floors of earlier versions with a single visible convention.

5. Prerequisites

Earlier versions carried 14 core gates whose failure capped the overall stage. In the Egypt assessment, 10 of the 14 carried no recorded value: the machinery with the most power over the result rested on the least evidence. Version 1.7 narrows the power and demands the evidence.

- Three universal prerequisites: rural mobile coverage (2.1), rural electricity (2.9), and a data protection law (4.1). Absence of any one blocks every use case column.
- Seven use case prerequisites, each blocking only its own column: farmer registry (3.3) and interoperability standards (3.11); agricultural data governance (4.5); digital ID (4.7); digital extension capability (5.5); agri-fintech rails (6.14); and consent and data rights safeguards for AI-enabled services (7.12).
- Two delivery flags, blocking nothing but shown on the report cover: an inter-ministerial coordination mechanism (4.9) and a ministry digital unit (5.7).

Prerequisites bind on presence only, the first rung of the ladder. That a country has no data protection law is a fact, and a fact may legitimately cap the ambition of a roadmap. That a country's data protection law is of middling quality is a judgment, and a judgment must never carry blocking power. A prerequisite whose presence cannot be evidenced shows as unverified: it cannot silently pass, and it cannot silently fail.

Two binding rules were settled in the August 2026 test runs and are carried here for ratification. First, the consent-and-data-rights prerequisite (7.12) is written for “all AI-enabled services”, a phrase that named no column and therefore bound nothing; it now gates the agricultural intelligence column. Second, a universal prerequisite recorded as present but narrow caps every use case column at partial. Absence blocks, but narrow presence must not read as readiness: without this rule Nigeria returned advisory and smart farming as ready on rural electrification of 23.5 percent.

6. The Assessment Process

The process has four steps. Two belong to people; two are automatic. A first pass is a matter of days, not months: 19 of the 57 indicators draw on global series that a machine can fetch, and the remaining 38 are country work that proceeds source by source.

The instrument is the scoring workbook, distributed as three files: a Blank Template for new assessments, and the Egypt and Nigeria worked examples. Each file is self-contained, with the same seven sheets: Read Me, which carries the legend and the gate sign-off block; Config, every constant beside its meaning; the shared Ladder; the Tiers sheet, which states what each source tier may carry; the Scoring sheet; an Issues sheet, the quality-control log; and a Visuals sheet of six live charts. An assessment touches exactly six entry columns on the Scoring sheet — value, source, source URL, tier, year and, for ladder indicators only, the assessor level. Everything else, including every chart, derives by formula.

Table 6.1 The four steps

Step	Who	Typical effort	Output
1.	Machine first, then	Machine pass: 1–2	57 indicators each carrying value,

Step	Who	Typical effort	Output
Assemble	assessor	days. Assessor pass: 1–2 weeks	source, and year, or a recorded gap
2. Derive	Machine, automatic	Immediate	Classes, levels, bands, layer profile, prerequisite status, flags
3. Report	Machine, automatic	Immediate	The diagnostic report (section 7), the model's single output
4. Use	TTL and team	Continuous through DAR preparation	Validated judgments; verify list closed by mission; refresh list executed; report feeds Playbook steps 1C, 2B, and 3C

Step 1: Assemble the evidence

The machine pass populates the measured indicators from global series (the WDI, the EGDI, the GCI, the GII, and their peers) and proposes documented readings from public artifacts. The assessor pass then does the country work: confirming artifacts, adding national statistics, scoring ladder indicators, and, importantly, recording gaps. An assessor who searches for a post-harvest loss figure and finds none has produced evidence, and the model treats it as such. The one rule that governs the whole step: no level without a recorded value. Table 6.2 shows the three kinds of entry.

Table 6.2 Example entries: a measurement, an artifact, and a recorded gap

ID	Value (as entered)	Source and year	Derived class	Derived level
2.4	41.2	World Bank WDI, 2024	Measured	3 (thresholds)
4.1	NDPA 2023 in force; NDPC enforcing	NDPC; Jones Day, 2025	Documented	4 (ladder)
1.5	DATA GAP — searched FAO, FMAFS, NSO	Machine pass, 2026	Gap	— (verify list)

Source hunting in the assessor pass follows the Source-Tier Protocol, a one-page companion to the specification. Official statistics and international databases come first (T1), then peer-reviewed and flagship work (T2), government artifacts (T3), and reputable grey literature (T4); news and vendor material (T5) is admissible only in the initiative register, and only for existence facts. Every citation is a deep link with an access date; a load-bearing number needs two independent T1 or T2 confirmations or ships flagged as single-source. The tier travels with the entry and is reported beside the evidence class, never weighted.

Step 2: Derive the scores

Everything computational is automatic and deterministic. The evidence class of each reading derives from what was recorded; levels derive from thresholds or the ladder; pillar bands, the layer profile, staleness flags, prerequisite status, and the use case readiness matrix all follow. There are no weights to set, no coverage percentages to compute, and no stage to deliberate. An assessor cannot tune the result, which is the point.

Step 3: Render the diagnostic report

The report described in section 7 is generated from the assessment without further human input. It is the endpoint of the model: a single document in which the indicators, their evidence, the visualizations, and the diagnostic narrative travel together. The workbook and its register are the instrument; the report is the deliverable. The workbook's Visuals sheet carries the same six charts live during scoring, so the team sees the profile form as the evidence comes in rather than at the end.

The rendered report carries two registers, visibly distinct. Scored sections render only what the engine computed from recorded values. Researched narrative, such as the problem statement's prose and the register's landscape reading, is set apart typographically, carries tier-badged citations, and may cite scored rows without ever setting a level. Two of its sections live outside the workbook: the initiative and solutions register, maintained as a companion file and admissible as Documented evidence for innovation and institutional indicators, and the strategic questions, derived mechanically from blocked cells, unverified prerequisites, and flagged constraints, then curated by the task team leader.

Step 4: Use the report in roadmap preparation

The team takes the report into the Playbook process: the pillar and layer profiles inform step 1C; the readiness matrix and the agriculture pillar inform the readiness and impact sides of step 2B; the KPI baseline seeds the step 3C results framework. Judged readings are validated with counterparts, the verify list becomes the mission agenda, and the refresh list is executed before drafting. The mandate question sits where the Playbook puts it: the diagnostic may be prepared at any time on public evidence, and the roadmap proper is a government-owned exercise.

Quality control is layered rather than appended, and it is specified in the one-page companion DAMM-v1.7-QC-Protocol.md. The rules of evidence in section 4 are the structural layer: they make classes derived, prerequisites factual, and weak evidence visible by construction. The render pipeline adds an automated layer of consistency, provenance, and reconciliation checks that must pass before a report is emitted; the result is stated in the report's footer. Three human gates complete the protocol: the assessor confirms every machine-filled row; a second reviewer checks all prerequisite and judged rows, plus a sample of the rest, before the first internal render; and the task team leader signs off against the four prohibitions before anything circulates — with findings recorded in an issues log inside the workbook.

7. Reading the Diagnostic Report

The report opens with an unnumbered executive brief, four tiles and the profile sentence, then runs ten sections, each answering one question and each named to the Playbook step it feeds.

Table 7.1 The ten sections of the diagnostic report

§	Section	The question it answers	Feeds
1	Country problem statement	What agricultural constraints must digital answer?	Playbook 1A
2	Initiative and solutions register	What exists already, at what scale,	Playbook 1B

§	Section	The question it answers	Feeds
		with what results, overlapping where?	
3	Pillar profile with evidence composition	Where is capability, and how much of it is actually evidenced?	Playbook 1C
4	Layer profile and structural gap	Are services running ahead of their rails?	Playbook 1C
5	Use case readiness matrix	Which use cases is the country ready for, and what blocks the rest?	Playbook 2B
6	Impact side	Which use case areas answer which agricultural constraints?	Playbook 2B
7	Binding constraints	What are the lowest-level indicators, prerequisites flagged?	Playbook 2B and 4A
8	Strategic questions	What must step 2 decide, stated as questions with their derivations?	Step 2 agenda
9	Evidence ledger: verify and refresh lists, vintage	What must the mission close, and what has aged out?	Mission planning
10	KPI baseline	What measured baselines exist, with source and year?	Playbook 3C

The executive brief that opens the report is unnumbered: four tiles and the profile sentence, so a reader who stops after one page still leaves with the finding and its evidence caveat. The act-or-verify reading of earlier drafts survives inside sections 7 and 9: a weak indicator that is measured is an investment case; a weak indicator that is judged or gap is a research task first.

Two kinds of outside context appear in the report, both bounded. A measured indicator may carry a reference value — the same global series read for the world or a peer set — beside the country value; references are context, never scored. And a strategic question may carry one precedent pointer: a tier-badged existence proof that the blocker it names has been cleared somewhere else. Pointers are drawn from the Global Practice Library (DAMM-v1.7-Practice-Library-Schema.md), a standing companion keyed to use case areas and prerequisite patterns; the readiness matrix, not narrative judgment, selects what is relevant, and the full practice review belongs to step 2 of the Playbook, not to the diagnostic.

7.1 Reading the visuals

Both the report and each workbook's Visuals sheet depict the same assessment six ways: a spider (radar) chart of the pillar profile; a bar chart of pillar means with values; a stacked bar of the evidence mix per pillar; a column chart of the four layers; a distribution of indicators by level; and the vintage of the evidence by observation year. The rule for all of them: read a level chart and the evidence-mix chart together. A strong bar standing on a mostly grey evidence bar is a hypothesis, not a finding.

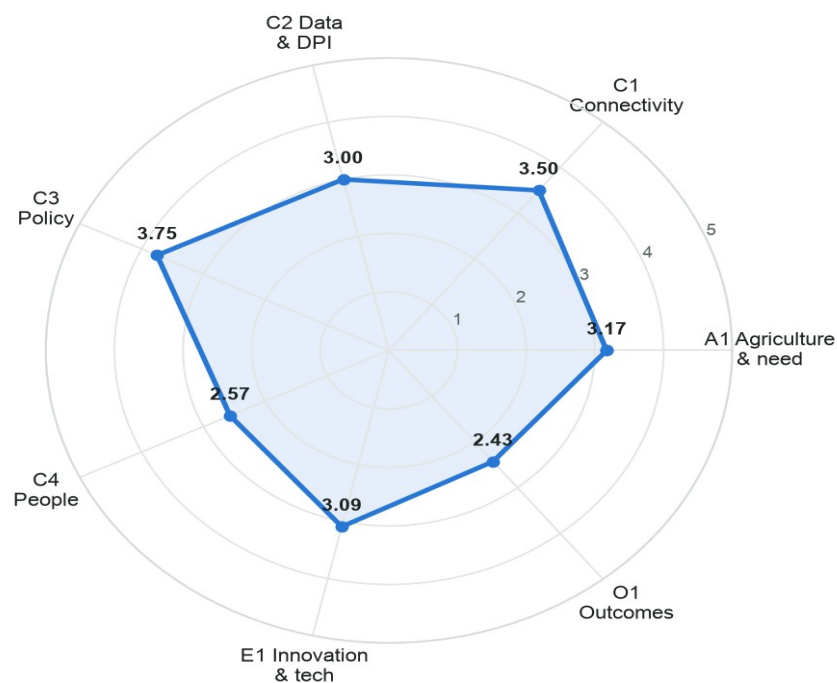


Figure 7.1 Egypt: pillar profile, mean level per pillar (1–5). Source: DAMM v1.6 Egypt workbook, Visuals sheet, August 2026.

The spider view is the fastest read of shape: Egypt’s polygon leans toward connectivity and policy and pulls inward at people and outcomes, which is the whole diagnostic in one glance. The evidence-mix chart is its conscience. Figure 7.2 shows Nigeria’s: the grey segments are recorded gaps, and their share is why three of Nigeria’s bands print in parentheses.

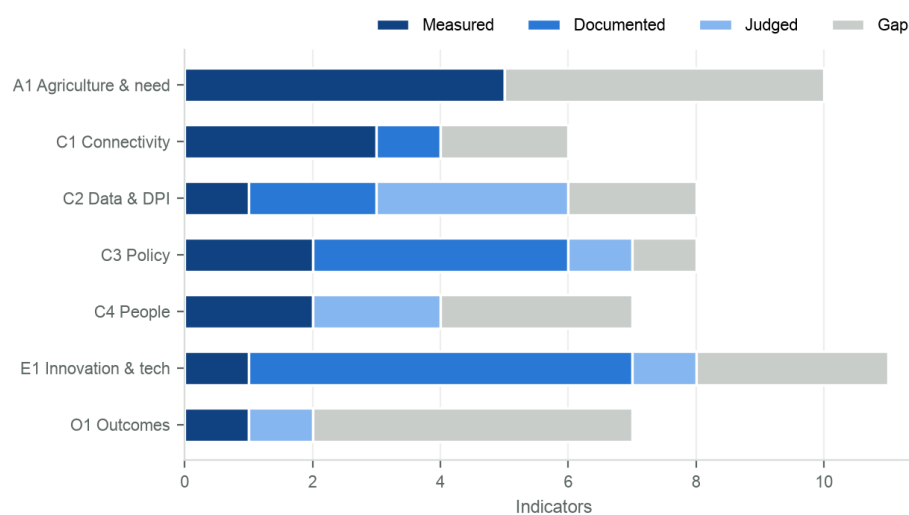


Figure 7.2 Nigeria: evidence mix per pillar, count of indicators by derived class. Source: DAMM v1.6 Nigeria workbook, Visuals sheet, August 2026.

Two reading conventions matter. A parenthesized band means the evidence behind it is mostly judgment or gap; treat it as a hypothesis. And the readiness matrix, not the layer gap, is the sharper

structural instrument: in the Nigeria test the national foundation-to-transformation gap read 0.0 while the matrix showed the agricultural intelligence column blocked by two absent prerequisites.

8. What the Model Will Not Do

Four prohibitions stand, unchanged in substance since version 1.3, and apply to every output of the model.

- No cross-country ranking. Bands are diagnostic categories, not league table positions, and reports are prepared one country at a time.
- No band, level, or profile may serve as a project development objective indicator, a disbursement-linked indicator, or a disbursement condition.
- No automatic financing, procurement, vendor, or technology decision may be derived from the diagnostic.
- No result may be claimed publicly before human review by the TTL and the responsible governance body.

It is important to point out that these prohibitions are design features rather than disclaimers. The absence of an overall score is what makes a league table impossible to construct; the derived evidence classes are what make a judgment impossible to launder into a measurement.

9. Country Applications

Version 1.7 was assessed against two countries in August 2026. Both runs used the same process end to end — a machine pass over the global series, then tiered source research for the remainder, then assessor confirmation and an independent adversarial review — so that any difference between them is a finding about the country and not an artifact of how each was assessed. Consistent with the first prohibition, the results are summarized separately and are not comparable.

Egypt. The assessment returned 20 measured readings, 32 documented, no judgments and 5 recorded gaps. Connectivity reads advanced (3.80); agriculture, data and DPI, policy, people, innovation and outcomes all read established. The need profile inverts the usual reading: cereal yields near twice the world average on land that is 94 percent equipped for irrigation, with the exposure sitting instead in a cereal import dependency of 43 percent and in inclusion. Agricultural intelligence is blocked, by three absent prerequisites: interoperability standards, an agricultural data governance framework, and agriculture-specific consent and data-rights safeguards. The other five columns return neither ready nor partial but unverified, because rural mobile broadband coverage — a universal prerequisite — is not published for Egypt by the regulator, the ministry or the ITU. The national figure is 99.8 percent, and recording it against a rural indicator would have returned five ready columns on evidence that does not measure the thing named. Eleven readings are stale and carry a dated refresh list.

Nigeria. The assessment returned 19 measured readings, 33 documented, 1 judgment and 4 recorded gaps. Outcomes read established (3.25), connectivity and policy established, and agriculture, data and DPI, people and innovation emerging. The structure is legible: a data protection act in force with active enforcement and a national AI strategy naming agriculture, beside

a digital agriculture strategy that has remained a draft since 2020, no agricultural data interoperability standards, and a national farmer registry that reaches under 1 percent of farm households. Agricultural intelligence is blocked. Every other column reads partial rather than ready, and at root for the same reason: rural electrification of 23.5 percent is present, but narrowly, and a universal prerequisite in that state caps the whole matrix.

Three methodological conclusions follow. The first is that depth of search, not the country, drove most of what a shallower pass had recorded as missing data: an assessment that had reported 21 gaps for Nigeria returned 4 once the tiered protocol was followed to its end, and the Egypt figure moved in the opposite direction, from an apparent zero to an honest 5, once migrated evidence was re-researched rather than inherited. The second is that the gaps which survive are largely not country facts at all: three of the four Nigerian gaps are also Egyptian gaps — the gender balance of the digital agriculture workforce, climate advisory reach, and a business-environment score whose assessment cohort has not yet published. An indicator that no country can evidence is a question for the register, and it is recorded as such in the ratification review. The third is that an indicator's name must be held to the evidence recorded against it. A separate audit of all 57 rows found six that named a construct their evidence did not measure; where the level turned on that difference the level was withdrawn, and the row now carries its evidence, its open question, and no score. Egypt's five unverified columns are the visible consequence.

Each country workbook's Visuals sheet carries the full chart set for that assessment, and the diagnostic reports embed the same views beside the narrative. On rich evidence the model orders priorities and dates them; on thin evidence it converts ignorance into a costed research agenda rather than a false score. The second behavior is the one earlier versions could not produce.

10. Limits of the Current Version

Four limits should be understood by anyone applying version 1.7, and each is scheduled for resolution in the ratification review.

- The A1 thresholds are test values. They produced sensible readings in both test countries, but they have not been ratified against sector benchmarks.
- The merge decisions that reduced 102 indicators to 57 are evidenced by two country assessments. Two is enough to justify the test, not to settle it; each further assessment either confirms a merge or argues to reopen it.
- The majority rule for weak-evidence bands treats an even split as sufficient evidence. Neither August 2026 assessment produced a tie once the evidence was deepened, so the rule stands untested in practice; a tie arguably should print as weak.
- The readiness matrix judges a column on prerequisites and mean levels. The narrow-presence rule now catches the worst case, but demand-side weakness still does not speak in the cell logic: Egypt reads five columns ready while farmer digital literacy is recorded only as an unrated document, no national rate being published. Demand-side indicators may deserve explicit weight.

Lastly, it is important to mention the register itself. Fifty-seven indicators is a floor as well as a ceiling: an indicator may be added only by naming the use case area and roadmap decision it serves,

and the census that records every such decision, with the evidence behind it, is maintained alongside the workbook.

Annex A. The Indicator Register

Fifty-seven indicators. "Absorbs" lists the version 1.5 indicators merged into the row, whose readings are retained as sub-readings. Method: T = numeric threshold, L = shared ladder. Source: G = global series (machine-fetchable), C = country sources.

ID	Indicator	Layer	Meth.	Src	Use cases	Absorbs
A1 · Agriculture and need						
1.1	Agriculture value added per worker (constant 2015 USD)	Foundation	T	G	NEED	—
1.2	Cereal yield (kg per hectare)	Foundation	T	G	NEED	—
1.3	Employment in agriculture (% of total employment)	Foundation	T	G	NEED	—
1.4	Food production index (2014-2016 = 100)	Foundation	T	G	NEED	—
1.5	Post-harvest loss rate (%)	Foundation	T	C	NEED	—
1.6	Smallholder access to formal markets (%)	Foundation	T	C	NEED	—
1.7	Agricultural credit access (% of farmers)	Foundation	T	C	NEED	—
1.8	Farmers using climate-smart/sustainable practices (%)	Foundation	T	C	NEED	—
8.1	Prevalence of undernourishment (% of population)	Foundation	T	G	NEED	—
8.5	Women who own land (% of agricultural holders)	Foundation	T	G	NEED, EQ	—
C1 · Connectivity and access						
2.1	Rural population covered by at least 3G (%) ♦	Foundation	T	G	ALL	2.2
2.4	Individuals using the Internet (% of population)	Foundation	T	G	ALL	2.3
2.5	Mobile broadband basket price (% GNI per capita)	Foundation	T	G	ALL	—
2.7	Smartphone ownership among rural/agricultural population (%)	Foundation	T	C	ALL	2.6
2.9	Electricity access in rural areas (%) ♦	Foundation	T	C	ALL	—
2.11	Device financing/subsidy schemes	Foundation	L	C	ALL	—

ID	Indicator	Layer	Meth.	Src	Use cases	Absorbs
	for farmers					
C2 · Data and DPI						
3.1	UN E-Government Development Index (EGDI)	Enablers	T	G	AGI, ALL	—
3.3	National farmer registry/database ♦	Enablers	L	C	FIN, AGI, ADV	—
3.4	Digital land/plot registration system	Enablers	L	C	FIN, AGI	—
3.5	National agricultural data portal / open data	Enablers	L	C	AGI, ADV, SMF	3.12, 3.13
3.6	Weather/climate data infrastructure for agriculture	Enablers	L	C	ADV, SMF, AGI	—
3.7	Satellite/earth observation data integration capability	Enablers	L	C	AGI, SMF	—
3.8	National soil map/database (digital)	Enablers	L	C	ADV, SMF	—
3.11	Interoperability standards for agricultural data ♦	Enablers	L	C	AGI, ALL	—
C3 · Policy and safeguards						
4.1	Data protection/privacy law ♦	Enablers	L	G	ALL	—
4.2	Cybersecurity legal framework (ITU GCI)	Enablers	T	G	ALL	—
4.3	Government AI Readiness Index	Enablers	T	G	AGI, ALL	—
4.4	National digital agriculture strategy	Enablers	L	C	ALL	—
4.5	Agricultural data governance framework ♦	Enablers	L	C	AGI, FIN	4.8
4.6	National AI strategy with agriculture component	Enablers	L	C	AGI, ALL	4.13
4.7	Digital ID system coverage (% of population) ♦	Enablers	T	C	FIN, ALL	—
4.9	Inter-ministerial coordination mechanism for digital agriculture ♦	Enablers	L	C	ALL	—
C4 · People and institutions						
5.2	Adult literacy rate (% age 15+)	Enablers	T	G	ALL	—
5.3	Tertiary enrollment in STEM fields (% gross)	Enablers	T	G	ALL	—
5.4	Digital literacy among farmers (%)	Enablers	T	C	ALL	—

ID	Indicator	Layer	Meth.	Src	Use cases	Absorbs
5.5	Extension workers trained in digital tools (%) ♦	Enablers	L	C	ADV	1.9, 5.6
5.7	Dedicated digital/AI units in Ministry of Agriculture ♦	Enablers	L	C	ALL	5.11
5.8	University programs in agtech/digital agriculture	Enablers	L	C	ALL	5.9
5.12	Gender balance in digital agriculture workforce	Enablers	T	C	EQ, ADV	5.13
E1 · Innovation, solutions, and emerging technology						
3.9	Digital advisory platforms reaching >10,000 users	Transformation	L	C	ADV	—
3.10	Digital agricultural market/e-commerce platforms (>5,000 active users)	Transformation	L	C	MKT	—
6.1	Global Innovation Index (GII) score	Enablers	T	G	ALL	—
6.3	Business Ready (B-READY) score	Enablers	T	G	ALL	—
6.4	Active agtech startups/firms	Transformation	L	C	ALL	6.5, 6.6, 6.10
6.9	Public-private partnerships in digital agriculture	Transformation	L	C	ADV, AGI, SCM	—
6.12	Digital public goods adopted in agriculture	Transformation	L	C	ALL	—
6.13	SME access to digital agriculture tools/platforms (%)	Transformation	T	C	SMF, SCM, MKT	6.7
6.14	Agri-fintech / mobile-money rails accessible to smallholders ♦	Transformation	L	C	FIN	6.11
7.2	AI systems deployed in public agricultural sector	Transformation	L	C	ALL-AI	7.3, 7.4, 7.5, 7.6, 7.7
7.12	Farmer consent and data rights in AI systems ♦	Enablers	L	C	ALL-AI	7.10, 7.11
O1 · Outcomes and inclusion						
8.2	Account ownership, female (% age 15+)	Outcomes	T	G	FIN, EQ	8.3
8.4	Mobile money account (% age 15+)	Outcomes	T	G	FIN	—
8.6	Gender gap in mobile phone ownership (percentage points)	Outcomes	T	C	EQ	—

ID	Indicator	Layer	Meth.	Src	Use cases	Absorbs
8.9	Smallholder farmers reached by digital services (% of total)	Outcomes	T	C	ALL, EQ	8.7, 8.8, 8.16
8.11	Services available in local languages (% of major platforms)	Outcomes	T	C	ADV, EQ	—
8.12	Documented yield improvement from digital tools (%)	Outcomes	L	C	ALL	8.13
8.17	Climate advisory reach (% farmers receiving actionable seasonal advisories)	Outcomes	T	C	ADV	8.14

◆ prerequisite indicator (three universal, seven use case, two delivery flags; section 4).

Annex B. Scoring Reference

B.1 Bands

Half-open, mutually exclusive; a pillar band is the unweighted mean of its indicators' levels.

Table B.1 Bands

Band	Range	Meaning of the boundary
Nascent	1.0 to below 1.8	Lower bound of the scale
Emerging	1.8 to below 2.6	2.6 is also the readiness threshold used by the matrix
Established	2.6 to below 3.4	
Advanced	3.4 to below 4.2	
Transformative	4.2 to 5.0	Upper bound of the scale

B.2 The ladder

Table B.2 The ladder

Level	Condition	Evidence required
1	Absent	The search that established absence
2	Announced	The announcement or draft
3	Adopted or operating narrowly	The instrument or system, with ownership
4	Operating with quality	Governance, funding, institutionalization
5	Operating at scale with evidence of use	Coverage against the indicator's scale measure

B.3 Rules and constants, with their meanings

- Staleness: observation year more than 3 years before the assessment year. The flag marks the reading for refresh; it does not remove the level.
- Weak-evidence rendering: judged plus gap readings outnumber measured plus documented in a pillar; the band prints in parentheses.
- Structural gap: mean level of the foundation layer minus mean level of the transformation layer. An absolute gap above 1.5 flags; a positive gap reads as unrealized foundations, a negative gap as services running ahead of their rails.
- Readiness: a use case column is blocked by an absent prerequisite (universal or its own), unverified if a prerequisite could not be evidenced, partial if a prerequisite is narrow or the mean level of its bearing indicators is below 2.6, and ready otherwise.
- Prerequisites bind on presence only. Quality and scale never block.

Annex C. Version History

Version	Substance
v1.0–v1.1	Initial framework, 8 pillars and 97 indicators; red-team corrections, live formulas, non-overlapping bands (2026).
v1.3–v1.4	Role families and three read-outs; core gates and coverage suppression; restored live scoring; 102 indicators; Egypt worked example (August 2026).
v1.5	Simplified vocabulary and process; diagnostic package sheet; scoring machinery unchanged (August 2026).
v1.6	Register cut 102 to 57 with a per-indicator census; agriculture scored; evidence classes derived from mandatory values; stages, read-outs, weights, confidence weights and coverage arithmetic removed; shared ladder; prerequisites bind on presence per use case; diagnostic report as the endpoint. Never circulated (August 2026).
v1.7	This version. Hardened by a clean-slate re-assessment of Egypt and Nigeria under one process, with assessor confirmation and independent review executed and twenty-eight defects logged. Evidence class made tier-aware; 7.12 bound to the agricultural intelligence column; universal prerequisites at narrow presence capping columns at partial; rounding aligned to the workbook; the standalone-report rule; workbooks rebuilt with source URL, tier, Tiers and Issues sheets and the gate sign-off block (August 2026).

References. World Bank, Gates Foundation, and BCG. 2025. Digital Agriculture Roadmap Playbook. · BCG for the Gates Foundation. 2020. DFS-SAID 2.0 Digital Agriculture Maturity Model. · World Bank. 2026. Bhutan DAR Assessment (March). · DAMM v1.6 Specification and Indicator Census, August 2026. · Egypt and Nigeria diagnostic reports, DAMM v1.6 test runs, August 2026. Country sources as cited in the reports: NDPC (2025), NIMC (2026), NITDA (2020), FMAFS (2025), ITU (2024), WIPO (2025), UN DESA (2024), World Bank WDI (2024–2025).